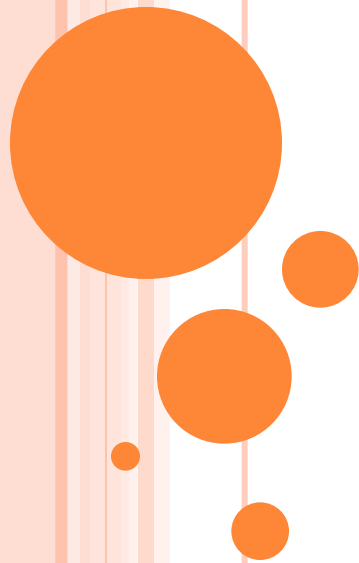


SIMPLIFICATION OF BOOLEAN FUNCTIONS

Veitch-Karnaugh diagrams



DEFINITIONS (I)

Definition 1: Let $f: (B_2)^n \rightarrow B_2$ be a Boolean function of n variables.

The set $S_f = \{(x_1, x_2, \dots, x_n) \mid f(x_1, x_2, \dots, x_n) = 1\}$, containing all the groups $(x_1, x_2, \dots, x_n) \in B_2^n$ for which f takes the value 1, is called the support of f .

Example: For $n=4$ variables, the support of the monom $m = x_1 \wedge \bar{x}_3$, is:

$$S_m = \{(1,1,0,1), (1,0,0,1), (1,1,0,0), (1,0,0,0)\}$$

Definition 2: The monom m is smaller or equal than the monom m' if and only if the support of m is included or equal to the support of m' : $m \leq m' \Leftrightarrow S_m \subseteq S_{m'}$

Example. Let $m = x_1 \wedge \bar{x}_2 \wedge x_4$ and $m' = x_1 \wedge \bar{x}_2$, be two monoms of $n=4$ variables.

$$S_m = \{(1,0,0,1), (1,0,1,1)\}, \quad S_{m'} = \{(1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1)\}$$

$$S_m \subseteq S_{m'} \text{ and thus } m \leq m', \quad x_1 \wedge \bar{x}_2 \wedge x_4 \leq x_1 \wedge \bar{x}_2.$$

Note: We use the concatenation of the variables to symbolize their conjunction.

Ex: the monom $m = x_1 \wedge \bar{x}_2 \wedge x_4$ is represented as $m = x_1 \bar{x}_2 x_4$.



DEFINITIONS (II)

Definition 3:

For a Boolean function of n variables let us consider the monoms:

$$m = x_{k_1}^{\alpha_{k_1}} \wedge \dots \wedge x_{k_j}^{\alpha_{k_j}} \wedge x_{k_i} \wedge x_{k_l}^{\alpha_{k_l}} \wedge \dots \wedge x_{k_s}^{\alpha_{k_s}} \text{ and}$$

$$m' = x_{k_1}^{\alpha_{k_1}} \wedge \dots \wedge x_{k_j}^{\alpha_{k_j}} \wedge \overline{x_{k_i}} \wedge x_{k_l}^{\alpha_{k_l}} \wedge \dots \wedge x_{k_s}^{\alpha_{k_s}}, \text{ where :}$$

$$k_1, \dots, k_j, k_i, k_l, \dots, k_s \in \{1, 2, \dots, n\} \text{ and } k_1 < \dots < k_j < k_i < k_l < \dots < k_s,$$

1. m and m' are called adjacent (neighboring) monoms because they differ only by the power of the variable with the index " k_i ".

Adjacency (neighborhood) relation is defined by a single variable change.

2. the factorization of the monoms m and m' is:

$$m \vee m' = x_{k_1}^{\alpha_{k_1}} \wedge \dots \wedge x_{k_j}^{\alpha_{k_j}} \wedge x_{k_l}^{\alpha_{k_l}} \wedge \dots \wedge x_{k_s}^{\alpha_{k_s}} \text{ obtained by eliminating the variable } "k_i".$$

Note that the result of factorization is a monom containing the common part of m and m' and this monom is bigger than both m and m' and it covers them.

FACTORIZATION - *EXAMPLES*

Minterms of 3 variables:

$$m_6 \vee m_2 = x_1^1 x_2^1 x_3^0 \vee x_1^0 x_2^1 x_3^0 = x_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 \bar{x}_3 = (x_1 \vee \bar{x}_1) x_2 \bar{x}_3 = 1 x_2 \bar{x}_3 = x_2 \bar{x}_3$$

$$m_5 \vee m_4 = x_1^1 x_2^0 x_3^1 \vee x_1^1 x_2^0 x_3^0 = x_1 \bar{x}_2 x_3 \vee x_1 \bar{x}_2 \bar{x}_3 = x_1 \bar{x}_2$$

Minterms of 4 variables:

$$m_{15} \vee m_{14} = x_1^1 x_2^1 x_3^1 x_4^1 \vee x_1^1 x_2^1 x_3^1 x_4^0 = x_1 x_2 x_3 x_4 \vee x_1 x_2 x_3 \bar{x}_4 = x_1 x_2 x_3 (x_4 \vee \bar{x}_4) = x_1 x_2 x_3 1 = x_1 x_2 x_3$$

$$m_0 \vee m_2 = x_1^0 x_2^0 x_3^0 x_4^0 \vee x_1^0 x_2^0 x_3^1 x_4^0 = \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 \bar{x}_2 x_3 \bar{x}_4 = \bar{x}_1 \bar{x}_2 \bar{x}_4$$



DEFINITIONS (III)

Definition 5: The set $M(f)$ is called the *set of maximal monoms* for the function $f: (B_2)^n \rightarrow B_2$ if:

1. $\forall m \in M(f), m \in FB(n), m \leq f$ and
2. $\forall m \in M(f), \exists m' \in FB(n)$ such that $m < m' \leq f$.

The maximal monoms are minterms or monoms obtained by using factorization.

Definition 6: The set $C(f)$ is called the *set of central monoms* for the function $f: (B_2)^n \rightarrow B_2$ if:

1. $\forall m \in C(f), m \in M(f)$ and
2. $\forall m \in C(f), m \not\leq \bigvee_{m' \in M(f) - \{m\}} m'$

Note: A maximal monom is a central monom for a function f , if the monom is not smaller than the disjunction of all the other maximal monoms of f .

Property: Let $f: (B_2)^n \rightarrow B_2$ be a Boolean function.

1. $M(f) \neq \emptyset$, but $C(f)$ can be empty.
2. $f(x_1, \dots, x_n) = \bigvee_{m \in M(f)} m$, but it is not always the simplest form of f .

SIMPLIFICATION OF BOOLEAN FUNCTIONS

Informally, to simplify a Boolean function given in a disjunctive canonical form, means to cover all its minterms with a minimum number of maximal monoms and with a minimum number of overlaps.

The simplification process is formalized by the steps below.

The steps 2, 3, 4 are specific to the applied simplification method.

1. The initial function f is transformed into $DCF(f)$.
2. Factorization process $\Rightarrow$ the set of maximal monoms $M(f)$.
3. From the set of maximal monoms the central monoms are selected $\Rightarrow C(f)$
4. The case of the simplification algorithm (presented below) is identified and all simplified forms are obtained.



SIMPLIFICATION ALGORITHM

Input data: f – a Boolean function in disjunctive canonical form

Output data: $f^s_1, f^s_2, \dots, f^s_k$ – all the disjunctive simplified forms of f

begin

Compute $M(f)$ and $C(f)$

if $M(f) = C(f)$ **then** // case 1--- one solution

$$f^s = \bigvee_{m \in M(f)} m ;$$

else

if $C(f) \neq \emptyset$ **then** // f has central monoms

// case 2 --- k solutions

$$g = \bigvee_{m \in C(f)} m$$

$f^s_i = g \vee h_i, i = \overline{1, k}, h_i$ is a disjunction of a minimum number of maximal monoms such that $S_{h_i} = S_f \setminus S_g$

else // $C(f) = \emptyset$, no central monoms

// case 3 --- k solutions

$f^s_i = h_i, i = \overline{1, k}, h_i$ is a disjunction of a minimum number of maximal monoms such that $S_{h_i} = S_f$

end_if

end_if

end_algorithm



SIMPLIFICATION METHODS

This algorithm is a general one and it is used in all three simplification methods:

- 1. Veitch-Karnaugh diagrams method – a graphical method**
- 2. Quine-Mc'Clusky's method – an analytical method**
- 3. Moasil's method – an algebraic method which uses the propositional logic.**



VEITCH-KARNAUGH DIAGRAMS METHOD

- it is applied to Boolean functions of 2, 3 and 4 variables
- the minterms from DCF are represented in the diagrams

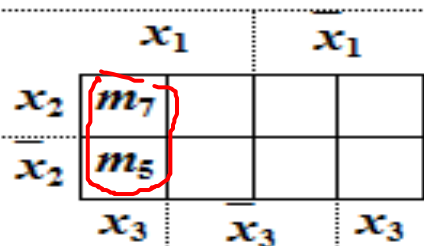
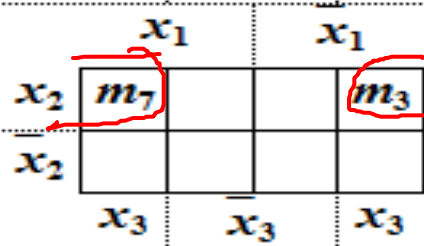
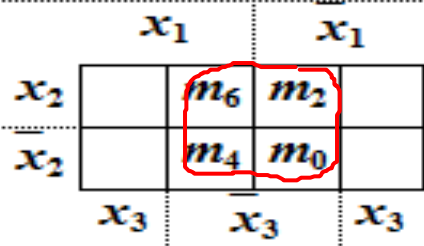
No. var.	Veitch diagram	Karnaugh diagram
2	$ \begin{array}{c} x_1 \quad \bar{x}_1 \\ x_2 \begin{array}{ c c } \hline m_3 & m_1 \\ \hline \end{array} \\ \bar{x}_2 \begin{array}{ c c } \hline m_2 & m_0 \\ \hline \end{array} \end{array} $	$ \begin{array}{c} x_1 \backslash x_2 \quad 0 \quad 1 \\ 0 \begin{array}{ c c } \hline m_0 & m_1 \\ \hline \end{array} \\ 1 \begin{array}{ c c } \hline m_2 & m_3 \\ \hline \end{array} \end{array} $
3	$ \begin{array}{c} x_1 \quad \bar{x}_1 \\ x_2 \begin{array}{ c c c c } \hline m_7 & m_6 & m_2 & m_3 \\ \hline \end{array} \\ \bar{x}_2 \begin{array}{ c c c c } \hline m_5 & m_4 & m_0 & m_1 \\ \hline \end{array} \\ x_3 \quad \bar{x}_3 \quad x_3 \end{array} $	$ \begin{array}{c} x_1 \backslash x_2 \backslash x_3 \quad 00 \quad 01 \quad 11 \quad 10 \\ 0 \begin{array}{ c c c c } \hline m_0 & m_1 & m_3 & m_2 \\ \hline \end{array} \\ 1 \begin{array}{ c c c c } \hline m_4 & m_5 & m_7 & m_6 \\ \hline \end{array} \end{array} $
4	$ \begin{array}{c} x_1 \quad \bar{x}_1 \\ x_2 \begin{array}{ c c c c } \hline m_{15} & m_{13} & m_5 & m_7 \\ \hline \end{array} x_4 \\ \bar{x}_2 \begin{array}{ c c c c } \hline m_{14} & m_{12} & m_4 & m_6 \\ \hline \end{array} \bar{x}_4 \\ x_3 \quad \bar{x}_3 \quad x_3 \end{array} $	$ \begin{array}{c} x_1 \backslash x_2 \backslash x_3 \backslash x_4 \quad 00 \quad 01 \quad 11 \quad 10 \\ 00 \begin{array}{ c c c c } \hline m_0 & m_1 & m_3 & m_2 \\ \hline \end{array} \\ 01 \begin{array}{ c c c c } \hline m_4 & m_5 & m_7 & m_6 \\ \hline \end{array} \\ 11 \begin{array}{ c c c c } \hline m_{12} & m_{13} & m_{15} & m_{14} \\ \hline \end{array} \\ 10 \begin{array}{ c c c c } \hline m_8 & m_9 & m_{11} & m_{10} \\ \hline \end{array} \end{array} $



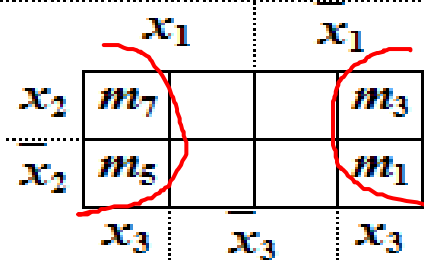
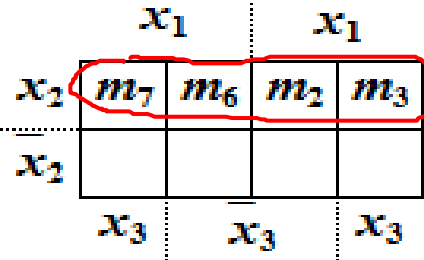
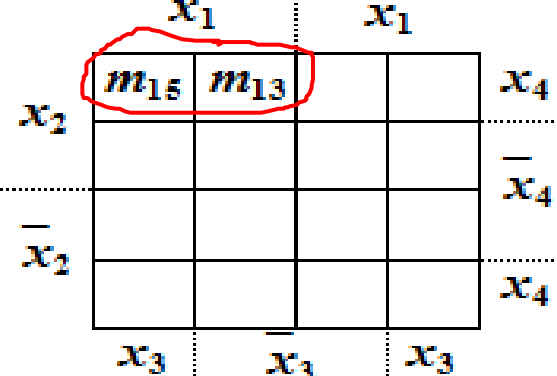
VEITCH-KARNAUGH DIAGRAMS METHOD

- The cells in a Veitch/Karnaugh diagram are arranged so that there is only a single variable change between neighbor (adjacent cells).
- Adjacency (graphical neighborhood) relation is defined by a single variable change.
- The diagrams are circular, meaning that the first row/column and the last row/column are neighbors.
- Two minterms belonging to two neighbor cells factorize $\Rightarrow$ *simple factorization*.
- The result of a *k-factorization* is a monom which contains the common variables of all 2^k neighbor minterms (cells).
- The maximal monoms are obtained from the diagram (Veitch/Karnaugh) by applying factorizations as follows: for a Boolean function of n variables we try first *n-factorization*, we continue with *n-1 factorizations*, ... , *simple factorizations* and *0-factorizations* (isolated minterms).

EXAMPLES OF FACTORIZATIONS (I)

Number of simplified variables	Veitch diagram	The maximal monom obtained after factorization
1 from 3 simple factorization		$m_7 \vee m_5 = x_1 x_3$
1 from 3 simple factorization		$m_7 \vee m_3 = x_2 x_3$
2 from 3 double factorization		$m_0 \vee m_2 \vee m_4 \vee m_6 = \overline{x_3}$

EXAMPLES OF FACTORIZATIONS (II)

2 from 3 double factorization		$m_1 \vee m_3 \vee m_5 \vee m_7 =$ $= x_3$
2 from 3 double factorization		$m_2 \vee m_3 \vee m_6 \vee m_7 =$ $= x_2$
1 from 4 simple factorization		$m_{13} \vee m_{15} = x_1 x_2 x_4$



EXAMPLES OF FACTORIZATIONS (III)



M. Lupea

Number of simplified variables	Veitch diagram	The maximal monom obtained after factorization
2 from 4 double factorization		$m_{12} \vee m_{13} \vee m_{14} \vee m_{15} = x_1 x_2$
2 from 4 double factorization		$m_1 \vee m_5 \vee m_9 \vee m_{13} = \bar{x}_3 x_4$



EXAMPLES OF FACTORIZATIONS (IV)

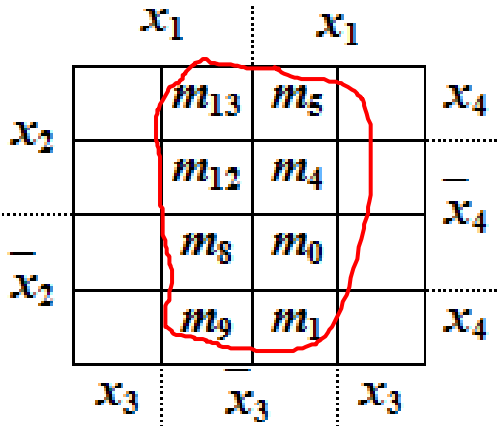
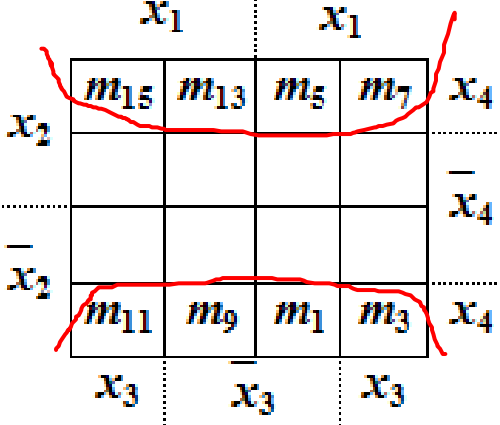


M. Lupea

Number of simplified variables	Veitch diagram	The maximal monom obtained after factorization
2 from 4 double factorization		$m_3 \vee m_7 \vee m_{11} \vee m_{15} =$ $= x_3 x_4$
2 from 4 double factorization		$m_{10} \vee m_{11} \vee m_{14} \vee m_{15} =$ $= x_1 x_3$



EXAMPLES OF FACTORIZATIONS (V)

Number of simplified variables	Veitch diagram	The maximal monom obtained after factorization
3 from 4 triple factorization		$m_0 \vee m_1 \vee m_4 \vee m_5 \vee$ $m_8 \vee m_9 \vee m_{12} \vee m_{13} =$ $= \bar{x}_3$
3 from 4 triple factorization		$m_1 \vee m_3 \vee m_5 \vee m_7 \vee$ $m_9 \vee m_{11} \vee m_{13} \vee m_{15} =$ $= x_4$



VEITCH-KARNAUGH DIAGRAMS METHOD

- Factorization process $\Rightarrow$ the set of *maximal monoms* $M(f)$.
- From the set of maximal monoms the *central monoms* are selected $\Rightarrow C(f)$
- The *central monoms* are the maximal monoms which cannot be covered by the disjunction of all the other maximal monoms.
- Identification of the *central monoms*:
 - if the group of the minterms covered by a maximal monom contains at least one cell (minterm) circled once, then the maximal monom is a central one and it belongs to all the simplified forms of the function.
- The groups of the minterms corresponding to the central monoms are shaded in the diagram.
- The minterms from the function's expression which are unshaded, are not covered by the central monoms and they will be covered in all the possible ways using a minimum number of unused maximal monoms and with a minimum number of overlaps, resulting *all the simplified forms* of the initial function.

EXAMPLE 1. SIMPLIFY THE BOOLEAN FUNCTION:

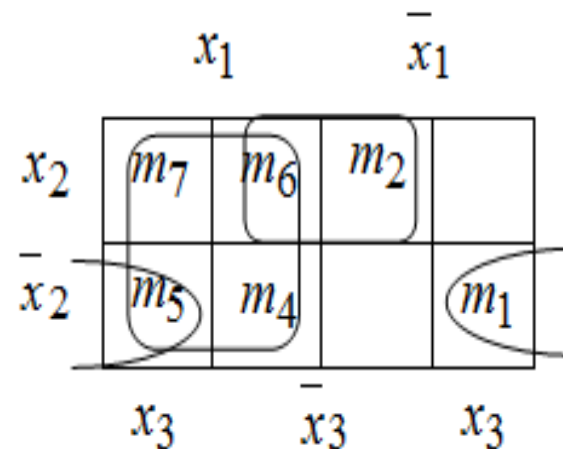
$$f(x_1, x_2, x_3) = x_1 x_2 x_3 \vee x_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 \bar{x}_3 \vee x_1 \bar{x}_2 x_3 \vee x_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3$$

$$= m_7 \vee m_6 \vee m_2 \vee m_5 \vee m_4 \vee m_1$$

Veitch diagram

	x_1	$\bar{x}_1$	
x_2	m_7	m_6	m_2
$\bar{x}_2$	m_5	m_4	m_1
	x_3	$\bar{x}_3$	x_3

Factorization process



- one group of 4 neighbor minterms
- two groups of 2 neighbor minterms

EXAMPLE 1 – FACTORIZATION PROCESS

double factorization

		x_1	$\bar{x}_1$	
x_2	m_7	m_6	m_2	
$\bar{x}_2$	m_5	m_4		m_1
	x_3	$\bar{x}_3$	$\bar{x}_3$	x_3

maximal monom: $max1 = m_7 \vee m_6 \vee m_5 \vee m_4 = x_1$

simple factorization

	x_1	$\bar{x}_1$	$\bar{x}_1$	
x_2	m_7	m_6	m_2	
$\bar{x}_2$	m_5	m_4		m_1
	x_3	$\bar{x}_3$	x_3	

maximal monom: $max3 = m_5 \vee m_1 = \bar{x}_2 x_3$

simple factorization

	x_1	x_1	$\bar{x}_1$	$\bar{x}_1$
x_2	m_7	m_6	m_2	
$\bar{x}_2$	m_5	m_4		m_1
	x_3	$\bar{x}_3$	$\bar{x}_3$	x_3

maximal monom: $max2 = m_6 \vee m_2 = x_2 \bar{x}_3$

The set of maximal monoms:

$$M(f) = \{max1, max2, max3\} = \\ = \{x_1, x_2 \bar{x}_3, \bar{x}_2 x_3\}$$



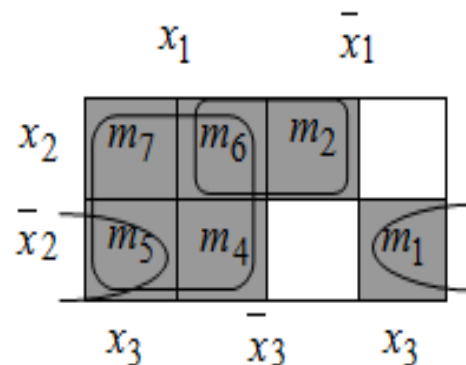
EXAMPLE 1(CONTD.)

Central monoms:

- a maximal monom is a central monom if its corresponding group of minterms contains at least one minterm circled once.

$$C(f) = \{max1, max2, max3\} = \{x_1, x_2 \bar{x}_3, \bar{x}_2 x_3\}$$

The minterms covered by the central monoms are shaded.



$M(f) = C(f)$ --- the first case of the simplification algorithm.

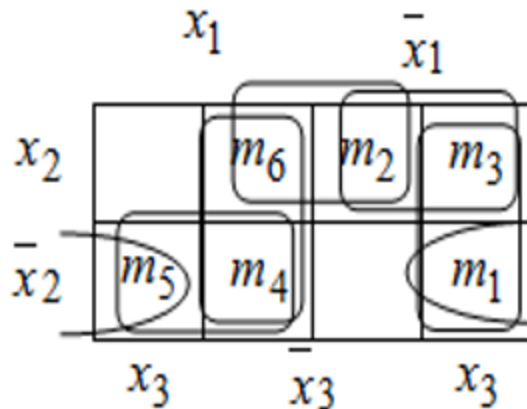
There is a unique simplified form of f , obtained as the disjunction of all central

monoms: $f^s(x_1, x_2, x_3) = max1 \vee max2 \vee max3 = x_1 \vee x_2 \bar{x}_3 \vee \bar{x}_2 x_3$

EXAMPLE 2. SIMPLIFY THE BOOLEAN FUNCTION:

$$\begin{aligned}
 f(x_1, x_2, x_3) &= x_2 \bar{x}_3 \vee \bar{x}_1 x_2 x_3 \vee x_1 \bar{x}_2 \vee \bar{x}_1 \bar{x}_2 x_3 = \\
 &= x_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 x_3 \vee x_1 \bar{x}_2 x_3 \vee x_1 x_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3 = \\
 &= m_6 \vee m_2 \vee m_3 \vee m_5 \vee m_4 \vee m_1
 \end{aligned}$$

Representation using Veitch diagram:



The set of *maximal monoms* is:

$$M(f) = \{max1, max2, max3, max4, max5, max6\}$$

$$max1 = x_1 \bar{x}_2 = m_5 \vee m_4, \quad max2 = x_1 \bar{x}_3 = m_4 \vee m_6$$

$$max3 = x_2 \bar{x}_3 = m_6 \vee m_2, \quad max4 = \bar{x}_2 x_3 = m_5 \vee m_1$$

$$max5 = \bar{x}_1 x_2 = m_2 \vee m_3, \quad max6 = \bar{x}_1 x_3 = m_3 \vee m_1$$

The set of *central monoms*:

$C(f) = \emptyset$ because all the minterms are circled twice.

$M(f) \neq C(f)$ and $C(f) = \emptyset$ — the third case of the simplification algorithm.

EXAMPLE 2- SIMPLIFIED FORMS

- All the minterms must be covered by a minimum number of maximal monoms, with a minimum number of overlaps.

$$\max 1 = x_1 \bar{x}_2 = m_5 \vee m_4, \quad \max 2 = x_1 \bar{x}_3 = m_4 \vee m_6, \quad \max 3 = x_2 \bar{x}_3 = m_6 \vee m_2,$$

$$\max 4 = \bar{x}_2 x_3 = m_5 \vee m_1, \quad \max 5 = \bar{x}_1 x_2 = m_2 \vee m_3, \quad \max 6 = \bar{x}_1 x_3 = m_3 \vee m_1$$

- Two simplified forms are obtained:

	x_1	$\bar{x}_1$	
x_2		m_6 m_2 m_3	
$\bar{x}_2$	m_5 m_4		m_1
	x_3	$\bar{x}_3$	x_3

$$\begin{aligned} f_1^s(x_1, x_2, x_3) &= \max 1 \vee \max 3 \vee \max 6 = \\ &= x_1 \bar{x}_2 \vee x_2 \bar{x}_3 \vee \bar{x}_1 x_3 \end{aligned}$$

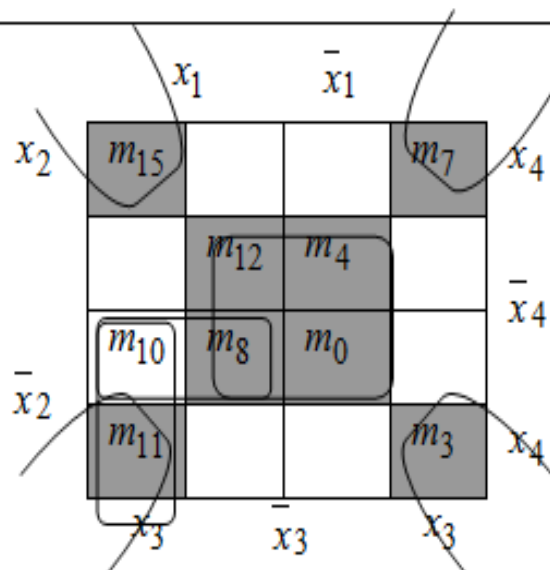
	x_1	$\bar{x}_1$	
x_2		m_6 m_2 m_3	
$\bar{x}_2$	m_5 m_4		m_1
	x_3	$\bar{x}_3$	x_3

$$\begin{aligned} f_2^s(x_1, x_2, x_3) &= \max 2 \vee \max 5 \vee \max 4 = \\ &= x_1 \bar{x}_3 \vee \bar{x}_1 x_2 \vee \bar{x}_2 x_3 \end{aligned}$$



EXAMPLE 3. SIMPLIFY THE BOOLEAN FUNCTION

$$\begin{aligned}
 f(x_1, x_2, x_3, x_4) &= x_1 x_2 x_3 x_4 \vee x_1 x_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 x_3 \bar{x}_4 \vee \\
 &\vee \bar{x}_1 x_2 x_3 x_4 \vee x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 x_3 x_4 \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 = \\
 &= m_{15} \vee m_{12} \vee m_4 \vee m_{10} \vee m_7 \vee m_8 \vee m_{11} \vee m_3 \vee m_0
 \end{aligned}$$



$$m_{15} \vee m_7 \vee m_{11} \vee m_3 = x_3 x_4 = \text{max1}$$

$$m_{12} \vee m_4 \vee m_0 \vee m_8 = \bar{x}_3 \bar{x}_4 = \text{max2}$$

$$m_{10} \vee m_{11} = x_1 \bar{x}_2 x_3 = \text{max3}$$

$$m_{10} \vee m_8 = x_1 \bar{x}_2 \bar{x}_4 = \text{max4}$$

$$M(f) = \{\text{max1}, \text{max2}, \text{max3}, \text{max4}\}$$

$C(f) = \{\text{max1}, \text{max2}\}$ – the set of central monoms

$M(f) \neq C(f)$ and $C(f) \neq \emptyset$ --- the second case of the simplification algorithm.

We denote by $g = \text{max1} \vee \text{max2} = x_3 x_4 \vee \bar{x}_3 \bar{x}_4$

The minterms covered by g are shaded in the diagram.

m_{10} is not covered by the central monoms and we can cover it in two ways: using max3 or using max4 .

Two simplified forms of f are obtained:

$$f_1^S(x_1, x_2, x_3, x_4) = g \vee \text{max3} = x_3 x_4 \vee \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 x_3$$

$$f_2^S(x_1, x_2, x_3, x_4) = g \vee \text{max4} = x_3 x_4 \vee \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 \bar{x}_4$$

EXAMPLE 4. SIMPLIFY THE BOOLEAN FUNCTION

f is given by its values 0:

$$f(0,0,0) = f(0,1,1) = f(1,1,1) = 0.$$

Canonical forms of f :

$$\begin{aligned} DCF(f) &= m_1 \vee m_2 \vee m_4 \vee m_5 \vee m_6 = \\ &= \bar{x}_1 \bar{x}_2 x_3 \vee \bar{x}_1 x_2 \bar{x}_3 \vee x_1 \bar{x}_2 \bar{x}_3 \vee x_1 x_2 \bar{x}_3 \vee x_1 x_2 x_3 \end{aligned}$$

$$\begin{aligned} CCF(f) &= M_0 \wedge M_3 \wedge M_7 = \\ &= (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \end{aligned}$$

Truth table of f :

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



EXAMPLE 4 (CONTD.)

Simplification method

$$\begin{aligned} DCF(f) &= m_1 \vee m_2 \vee m_4 \vee m_5 \vee m_6 = \\ &= \bar{x}_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 x_3 \vee x_1 \bar{x}_2 \bar{x}_3 \vee x_1 \bar{x}_2 x_3 \end{aligned}$$

Factorization process (Karnaugh diagram):

$x_1 \backslash x_2 x_3$	00	01	11	10
0		m_1		m_2
1	m_4	m_5		m_6

$$\begin{aligned} max_1 &= x_1 \bar{x}_2 = m_5 \vee m_4 & max_2 &= \bar{x}_1 x_3 = m_4 \vee m_6 \\ max_3 &= x_2 \bar{x}_3 = m_6 \vee m_2 & max_4 &= x_2 x_3 = m_5 \vee m_1 \end{aligned}$$

$M(f) = \{max_1, max_2, max_3, max_4\}$: **maximal monoms**

$C(f) = \{max_3, max_4\}$: **central monoms**

Second case of the *simplification algorithm*.

We denote by $g(x_1, x_2, x_3) = max_3 \vee max_4$

There are two **disjunctive simplified forms** of f .

$$\begin{aligned} f_1^{DS}(x_1, x_2, x_3) &= g \vee max_1 = max_3 \vee max_4 \vee max_1 = \\ &= x_2 \bar{x}_3 \vee x_2 x_3 \vee x_1 \bar{x}_2 \end{aligned}$$

$$\begin{aligned} f_2^{DS}(x_1, x_2, x_3) &= g \vee max_2 = max_3 \vee max_4 \vee max_2 = \\ &= x_2 \bar{x}_3 \vee x_2 x_3 \vee x_1 x_3. \end{aligned}$$

Dual simplification method

$$\begin{aligned} CCF(f) &= M_0 \wedge M_3 \wedge M_7 = \\ &= (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \end{aligned}$$

Dual factorization process (Karnaugh diagram):

$x_1 \backslash x_2 x_3$	00	01	11	10
0	M_0		M_3	
1			M_7	

$$\begin{aligned} maxd_1 &= M_3 \wedge M_7 = (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) = \\ &= (\bar{x}_1 \wedge \bar{x}_1) \vee \bar{x}_2 \vee \bar{x}_3 = \bar{x}_2 \vee \bar{x}_3 \\ &\text{(the common part of } M_3, M_7) \end{aligned}$$

$$maxd_2 = M_0 = x_1 \vee x_2 \vee x_3 \text{ (isolated maxterm)}$$

$Md(f) = \{maxd_1, maxd_2\}$: **maximal disjunctions**

$Cd(f) = Md(f)$: **central disjunctions**

First case of the *dual simplification algorithm*.

There is a unique **conjunctive simplified form** of f .

$$\begin{aligned} f^{CS}(x_1, x_2, x_3) &= maxd_1 \wedge maxd_2 = \\ &= (\bar{x}_2 \vee \bar{x}_3) \wedge (x_1 \vee x_2 \vee x_3). \end{aligned}$$

EXAMPLE 4 (CONTD.)

Let us compare the **disjunctive simplified forms** with the **conjunctive form**:

$$f_1^{DS}(x_1, x_2, x_3) = \overline{x_2}x_3 \vee x_2\overline{x_3} \vee x_1\overline{x_2}$$

$$f_2^{DS}(x_1, x_2, x_3) = \overline{x_2}x_3 \vee \overline{x_2}x_3 \vee x_1\overline{x_3}.$$

$$f^{CS}(x_1, x_2, x_3) = (\overline{x_2} \vee \overline{x_3}) \wedge (x_1 \vee x_2 \vee x_3).$$

- f_1^{DS} and f_2^{DS} have 6 occurrences of variables, 5 binary operations and 3 unary operations.
- f^{CS} has 5 occurrences of variables, 4 binary operations and 2 unary operations.

$f^{CS}(x_1, x_2, x_3) = (\overline{x_2} \vee \overline{x_3}) \wedge (x_1 \vee x_2 \vee x_3)$ is the simplest equivalent form of f .

